

from the same domain. The challenge is that N is small (18–30), which makes standard fine-tuning prone to memorization rather than learning generalizable features.

We address this by using a frontier VLM M_f (GPT-4.1) to generate rich visual reasoning for each training image. The frontier model does not classify the image. Instead, it receives the correct answer and explains why that answer is correct based on what it sees. This produces training data that teaches the small model how to reason about visual features, not just memorize image-label mappings. This approach is inspired by CoT distillation [9] and the SEAL framework [19], but differs in two key ways: we condition on the correct answer (unlike standard CoT distillation where the teacher must solve the task), and we operate in the vision-language domain (unlike SEAL which addresses text-only LLMs).

3.2 Prompt Design

Each task uses a single classification prompt that serves as both the training prompt and the evaluation prompt. This prompt contains: (1) image context (modality, resolution, magnification where relevant), (2) all class definitions with detailed visual descriptions, (3) contrastive features distinguishing similar classes, and (4) output format instruction (JSON).

The same prompt is used for both approaches (Direct and CoT-augmented) and for evaluation. This consistency is critical. In early experiments, using weaker class definitions during training degraded accuracy by over 18 percentage points, even with the same training images.

3.3 Answer-Conditioned CoT Generation

For each training image (x_i, y_i) , we generate justified visual descriptions from the frontier model. The key design choices are:

Answer-conditioning. The frontier model receives the correct label y_i along with the image x_i . It does not classify. It explains why y_i is correct based on what it observes. This ensures all training reasoning is directed toward the correct conclusion.

Without answer-conditioning, the frontier model would need to classify correctly to produce useful reasoning. Since its accuracy ranges from 29.6% to 91.1% across our tasks, a significant fraction of unconditioned reasoning would be wrong.

Contrastive reasoning. The generation prompt instructs the frontier model to explain not only why the image matches the given class, but also why it does not match the most similar alternative class. This teaches decision boundaries between confusable categories.

Multiple descriptions per image. We generate 3 descriptions per image at temperature 0.7, providing diversity in reasoning style while maintaining correctness. Combined with 1 direct JSON-only example, this yields 4 training pairs per image.

Prompt stripping for CoT examples. For the CoT training pairs, the “Respond with JSON” instruction is removed from the user prompt. This way, the model learns two response modes: when asked for JSON, output JSON directly; when asked without the JSON constraint, output reasoning followed by JSON.

3.4 Training Data Construction

For each task, we construct two training datasets from the same N labeled images: